

Proposed Solution

Date	01 July 2026
Team ID	SWTID-2026-2030
Project Name	Voice Based Concept Understanding Analyser
Maximum Marks	2 Marks

Proposed Solution:

S.No	Parameter	Description
1	Problem Statement (Problem to be solved)	Students often struggle to explain concepts clearly and do not receive instant feedback on their spoken explanations. Traditional evaluation methods are manual, time-consuming, and lack objective assessment of conceptual understanding and communication skills.
2	Idea / Solution Description	Develop an AI-powered Voice-Based Concept Understanding Analyzer (VBCUA) that records students' voice explanations, converts speech into text using Whisper, extracts audio features with Librosa, evaluates conceptual understanding using Sentence-BERT (SBERT), and provides instant feedback, performance scores, visualizations, and downloadable PDF reports.
3	Novelty / Uniqueness	The proposed solution combines speech recognition, semantic analysis, and AI-powered evaluation in a single platform. Unlike traditional assessment methods, it provides real-time personalized feedback, measures conceptual understanding objectively, and helps students improve both communication and subject knowledge.
4	Social Impact / Customer Satisfaction	The system enhances students' learning experience by providing immediate and accurate feedback, improving confidence, communication skills, and conceptual clarity. It also reduces teachers' evaluation workload and supports digital education through automated assessment.
5	Business Model (Revenue Model)	The application can follow a Software-as-a-Service (SaaS) model by offering subscription plans to educational institutions, colleges, coaching centers, and individual learners. Premium features such as advanced analytics, detailed reports, and institutional dashboards can generate additional revenue.
6	Scalability of the Solution	The solution is highly scalable and can support thousands of users through cloud deployment. It can be integrated with Learning Management Systems (LMS), online learning platforms, and educational institutions. Future enhancements include multilingual speech evaluation, interview assessment, and AI-based personalized learning recommendations.